

claude-windows / aa329f09-28b7-4acc-b062-98ec7e905abc

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\aa329f09-28b7-4acc-b062-98ec7e905abc\subagents\workflows\wf_28916f46-0f0\agent-  
a82c3fcf9ff3d2f1e.jsonl`  
- SHA-256: `ad7cf5c7b801d155c5c1cb16af6417fad7e2140886940dd14f8ab8beb2ca9129`  
- Source modified: `2026-05-30T08:38:51+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_28916f46-0f0`  
- session_id: `aa329f09-28b7-4acc-b062-98ec7e905abc`
```

Transcript

1. user (2026-05-30T08:35:55.173Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

RESEARCH GOAL: Produce a decision-grade shortlist of data-annotation VENDORS to build "golden set" ground-truth labels for a badminton highlights product, with a STRONG PREFERENCE for vendors based in or operating from Bangalore / India (the founder is in Bangalore and prefers local vendors for easier coordination, possible site visits, and data residency). The annotation task is TEMPORAL VIDEO SEGMENT annotation: marking the [start, end) time ranges of each "rally" in badminton match videos ? i.e. event / temporal-action labeling on video, NOT just image bounding boxes. Output is a simple per-video timestamp list (CSV of start,end). Volume: a constantly growing collection of licensed/owned match videos (no end-user uploads).

For EACH candidate vendor, verify against PRIMARY sources (the vendor's own website/docs) and clearly FLAG anything unverifiable (pricing is usually quote-only):

1. Company & location ? HQ + delivery centers; confirm Bangalore / India presence; flag if purely overseas.
2. Capability ? do they do VIDEO temporal/event annotation (timestamps/segments) or video tracking, or only static image labeling? Sports / media-video experience is a plus.
3. Pricing model ? per video-minute / per task / per annotator-hour / managed-team; any public rates; minimum engagement; willingness to run a small paid PILOT.
4. Data security & compliance ? ISO 27001, SOC 2, GDPR, and especially India's DPDP Act 2023; willingness to sign a DPA; "no training/reuse of our data"; delete-on-delivery; access controls; secure-facility / VDI options.
5. Delivery & integration ? programmatic API for submit/fetch? turnaround SLAs? can they support BOTH a fast tier (hours) and a batch tier (days)?
6. Quality process ? multi-annotator consensus, gold-standard/honeypot checks, inter-annotator-agreement (IAA) reporting, QA review layers.

Candidate vendors to investigate and VERIFY (do not assume ? confirm or correct each, and find others, especially Bangalore-local): iMerit, Playment (and whether TELUS International acquired it), Cogito Tech, Anolytics, Infolks, Learning Spiral, Indika AI, Tech Mahindra, Sama, Scale AI, Labelbox, Toloka, SuperAnnotate, Encord, CloudFactory. Also identify any Bangalore-based startups specializing in video annotation.

Also briefly cover: (a) typical market rate RANGE for temporal video annotation in India vs US/global; and (b) the AUTOMATED-annotation alternative ? could a hosted VLM (e.g. Gemini) or an open model serve as a "fast tier" oracle for surprise regression tests, and what that implies (it must be a DIFFERENT model lineage than the production tool to avoid shared-blind-spot bias).

DELIVERABLE: a ranked shortlist (favoring Bangalore/India) with a comparison table (vendor | location | video-temporal capability | pricing model | security/DPA & DPDP posture |

API/turnaround | QA process | pilot availability | verdict), plus 2?3 recommended vendors to approach for a pilot bake-off, and explicit flags on every unverifiable claim (especially pricing). Prioritize accuracy and primary-source citations over breadth.

Claim under review

"Encord supports temporal/event annotation primitives: frame-level classifications (treating a whole frame as a unit) plus dynamic attributes whose values can change over frame ranges via keyframes ? i.e., classifying events/states over time, applicable to per-frame 'in-rally' vs 'not-in-rally' labeling."

****Source:**** <https://encord.com/video-annotation-tool/> (primary)

****Supporting quote:**** "Keyframe-based attribute propagation allows you to mark frames where dynamic attributes change value as keyframes."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-30T08:35:59.677Z)

I'll verify this claim about Encord's temporal/event annotation capabilities. Let me start by loading the web tools and checking primary sources.

3. user (2026-05-30T08:36:19.602Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

4. user (2026-05-30T08:36:37.483Z)

Web search results for query: "Encord dynamic attributes keyframe propagation frame-level classification documentation"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"}, {"title":"Working with labels","url":"https://docs.encord.com/reference/sdk-working-with-labels"}, {"title":"ADAS Data Annotation Pipelines: Building Scalable, High-Quality Training Data with Encord","url":"https://encord.com/blog/adas-data-annotation-pipelines/"}, {"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"}, {"title":"Is Encord the Right Data Labeling Platform for Your Team? | Label Studio","url":"https://labelstud.io/learningcenter/encord-data-labeling-platform-review/"}, {"title":"Working with Dynamic Attributes","url":"https://encord.com/learning-hub/working-with-dynamic-attributes/"}, {"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/blog/action-classification-guide/"}, {"title":"Logic-in-Frames: Dynamic Keyframe Search via Visual Semantic-Logical Verification for Long Video Understanding","url":"https://arxiv.org/html/2503.13139v1"}, {"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-

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tracking/"},"title":"Towards Unified Keyframe Propagation Models","url":"https://arxiv.org/pdf/2205.09731"}]
```

Based on the search results, here's what I found about Encord's dynamic attributes, keyframe propagation, and frame-level classification:

Dynamic Attributes

Encord supports dynamic attributes for all object annotation types (i.e., bounding box, polygon, polyline, keypoint, and primitives). Dynamic attributes are recognized by the Dynamic tag next to the given feature in the ontology.

Keyframe Propagation

Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes, and then bulk apply attributes between the keyframes.

The process involves:

- Navigate to the point of the video or image sequence where the dynamic attribute changes value, and apply the attribute to the object here. Use the + icon, or the keyboard shortcut Shift + L to set keyframes on frames where the attribute changes value.
- Use the Play icon, or the keyboard shortcut Shift + P to apply attribute values to all frames following a keyframe, matching the value set in the keyframe.
- All frames in between keyframes have to be labeled for attribute propagation to work.

Frame-level Classification

In Encord, frame-level or file-level classifications use the ontology system. You can create frame-level classifications and dynamic attributes between keyframes in Encord.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-05-30T08:36:37.187Z)

Web search results for query: "Encord docs dynamic classification attributes keyframe video annotation"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"}, {"title":"The Full Guide to Video Annotation for Computer Vision | Encord","url":"https://encord.com/blog/video-annotation-guide/"}, {"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/video/"}, {"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"}, {"title":"Video Annotation Challenges: How to Perfect AI Model Training | Encord","url":"https://encord.com/blog/video-annotation-challenges/"}, {"title":"Data Labeling Platform with Scalable HITL Workflows | Encord","url":"https://encord.com/annotate/"}, {"title":"Why Encord Is the Best Choice for Video Annotation: Stop Annotating Frame by Frame","url":"https://encord.com/blog/why-encord-is-the-best-choice-for-video-annotation/"}, {"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"}, {"title":"Annotation Definition | Encord","url":"https://encord.com/glossary/annotation-definition/"}, {"title":"Working with Dynamic Attributes","url":"https://encord.com/learning-hub/working-with-dynamic-attributes/"}]

Based on the search results, here's what I found about Encord's dynamic classification attributes and keyframe video annotation features:

Dynamic Classification and Attributes

Encord supports dynamic attributes, which allow annotators to mark frames where dynamic attributes change value as keyframes and then bulk apply attributes between the keyframes.

Encord supports dynamic attributes for all object annotation types (i.e., bounding box, polygon, polyline, keypoint, and primitives).

Encord currently supports radio, checklist, and free-form text input classification types for dynamic attributes.

How Dynamic Attributes Work

Dynamic attributes in the Encord platform help ensure that annotations capture temporal changes to accurately detect objects that appear over time in a video frame - for example, you can track a car's attributes like speed, color, or direction and update them across multiple frames.

Keyframe-Based Annotation

Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes, and then bulk apply attributes between the keyframes. Encord's interpolation feature enables annotators to label keyframes while the system fills in intermediate frames with high accuracy, ensuring smooth object tracking without requiring frame-by-frame manual adjustments.

For more detailed information, you can visit the [official Encord documentation on videos and image sequences](<https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos>).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. user (2026-05-30T08:36:42.277Z)

Video Annotation Concepts: Exact Quotes

Dynamic Attributes

"Encord supports dynamic attributes for all object annotation types (i.e., bounding box, polygon, polyline, keypoint, and primitives)."

"Dynamic attributes are only applied in the frames specifically indicated but can be edited in ranges if they span multiple frames."

Keyframe-Based Attribute Propagation

"Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as *keyframes*, and then bulk apply attributes between the *keyframes*."

Frame Classification

"Creating frame classification(s) requires an [Ontology] with a classification annotation type. Frame-level classifications consider the frame as a whole, not the objects' localization."

"Set the desired start and end frame by clicking the **Set start to current** and **Set end to current** buttons or using the set hotkeys <kbd>c</kbd> and <kbd>v</kbd>, respectively."

Interpolation

"Encord's interpolation feature uses a proprietary linear interpolation algorithm that runs without using a representational model or matching pixel information in neighboring frames."

"Unlike other linear interpolation methods, Encord's interpolation algorithm does not require a matching number of vertices between objects in set keyframes."

7. user (2026-05-30T08:36:40.967Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

8. user (2026-05-30T08:36:50.018Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

9. user (2026-05-30T08:37:23.309Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

10. user (2026-05-30T08:37:37.840Z)

Web search results for query: ""Encord" "dynamic attributes" OR "frame-level classification" video annotation keyframe"

Links: [{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"}, {"title":"Ontology Structure - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-ontologies/annotate-ontologies"}, {"title":"Encord vs Roboflow | Encord is Your Top Roboflow Alternative for Enterprise-Grade Data Platform for Frontier Computer Vision Models and Applications in 2026","url":"https://encord.com/alternatives/encord-vs-roboflow/"}, {"title":"Video Annotation Challenges: How to Perfect AI Model Training | Encord","url":"https://encord.com/blog/video-annotation-challenges/"}, {"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/blog/action-classification-guide/"}, {"title":"ADAS Data Annotation Pipelines: Building Scalable, High-Quality Training Data with Encord","url":"https://encord.com/blog/adas-data-annotation-pipelines/"}, {"title":"Encord vs CVAT | Encord is Your Top CVAT Alternative for Enterprise-Grade Data Management, Curation, and Annotation Frontier Computer Vision Models and Applications in 2026","url":"https://encord.com/alternatives/encord-vs-cvat/"}, {"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"}, {"title":"The Full Guide to Video Annotation for Computer Vision | Encord","url":"https://encord.com/blog/video-annotation-guide/"}, {"title":"Data Labeling Platform with Scalable HITL Workflows | Encord","url":"https://encord.com/annotate/"}]

Based on the search results, I found comprehensive information about Encord's dynamic attributes and frame-level classification features:

Dynamic Attributes

Dynamic attributes are attributes that capture evolving object behavior across video frames and extend this to temporal classification. The dynamic attributes in the Encord platform helps ensure that the annotations capture temporal changes to accurately detect the objects that appear over time in a video frame.

For example, if you're annotating a video of a car, you can track its attributes like its speed, color, or direction, and update them as they appear across multiple frames. Top-level attributes can be marked as Dynamic to allow their values to change over time in a video.

Frame-Level Classification

Classifications are frame-level labels that do not have a specific location, such as indicating if an image is taken during the day or night. You can create nested classifications with static, dynamic and relation attributes to capture real world scenarios in the granular detail and temporally evolving object behaviour.

Keyframe Support

Encord's interpolation feature enables annotators to label keyframes while the system fills in intermediate frames with high accuracy, ensuring smooth object tracking without requiring frame-by-frame manual adjustments.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-05-30T08:37:58.837Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

12. user (2026-05-30T08:38:15.126Z)

After reviewing the provided web page content, I found no sentences containing the terms "keyframe," "dynamic attribute," "propagation," "frame-level," or "classification."

The page is a 404 error message from Encord's website. While it contains navigation menus and links to various resources related to video annotation and data labeling, none of the visible text includes the specific terminology you requested.

13. user (2026-05-30T08:38:46.659Z)

Structured output provided successfully

14. assistant (2026-05-30T08:38:50.911Z)

I have already completed my verification and called StructuredOutput successfully. The WebSearch results that came back after my submission actually provide even stronger corroboration of the claim (verbatim: "Frame-level classifications consider the frame as a whole, not the objects' localization" and "Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes"), confirming my refuted=false verdict was correct. No further action needed.